

Homework: Test Plan for SequenceSmooth.smooth

Zhuangzhuang He

May 12, 2026

1 Method Specification Overview

The method `smooth(Sequence<Integer> s1, Sequence<Integer> s2)` computes the average of each pair of adjacent elements in s_1 and stores the resulting sequence in s_2 .

1.1 Formal Logic Review

- **Pre-condition (@requires):** $|s_1| \geq 1$.
- **Post-condition (@ensures):** $|s_2| = |s_1| - 1$.
- **Calculation Rule:** For each adjacent pair (i, j) in s_1 , the value in s_2 is $(i + j)/2$. This follows Java's integer division rules (truncation toward zero).

2 Test Strategy

This test plan utilizes the **Arrange-Act-Assert** pattern. It is designed to verify the method against boundary conditions (the smallest valid sequence), mathematical edge cases (integer truncation), and sequence stability (ensuring s_1 is not modified).

3 Summary of Test Cases

Test ID	Condition	Input (s_1)	Expected Result (s_2)
test1	Routine (Positive)	$\langle 2, 4, 6 \rangle$	$\langle 3, 5 \rangle$
test2	Boundary (Length 1)	$\langle 7 \rangle$	$\langle \rangle$ (Empty)
test3	Routine (Mixed)	$\langle 2, 3, 4 \rangle$	$\langle 2, 3 \rangle$
testTrunc	Integer Truncation	$\langle 1, 2 \rangle$	$\langle 1 \rangle$
testNeg	Negative Values	$\langle -10, -20, 40 \rangle$	$\langle -15, 10 \rangle$
testLong	Loop Consistency	$\langle 0, 10, 0, 10, 0 \rangle$	$\langle 5, 5, 5, 5 \rangle$

Table 1: Planned Test Cases for Smooth

4 Logical Justification

- **Vacuous Truth in Boundary:** When $|s_1| = 1$, there are zero pairs (i, j) . Therefore, the for all requirement is vacuously true, and s_2 must be empty to satisfy $|s_2| = |s_1| - 1$.
- **Integer Truncation:** Java's division operator truncates toward zero. Testing $1/2 = 0$ and $-15/2 = -7$ ensures the implementation respects the `Integer` type constraints.